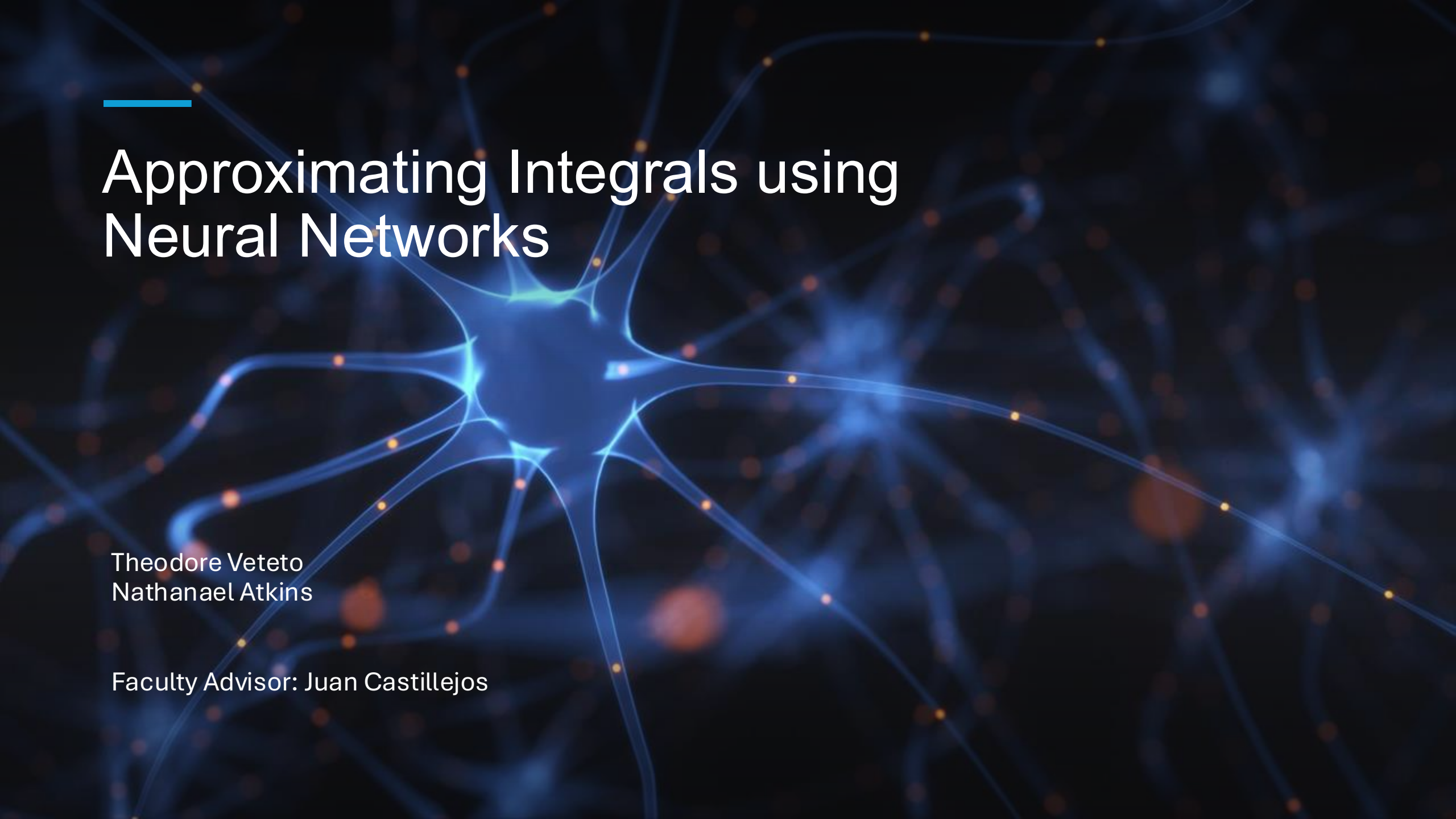


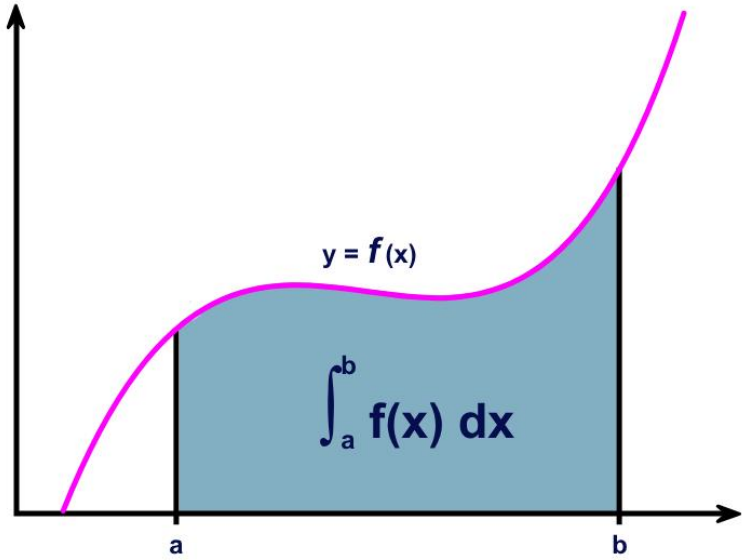
Approximating Integrals using Neural Networks

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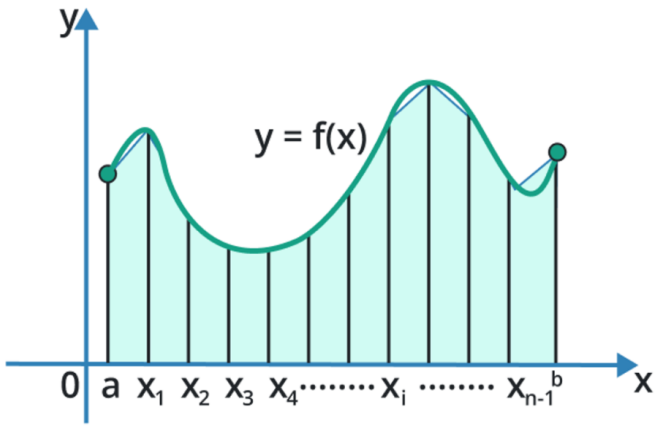
Integration

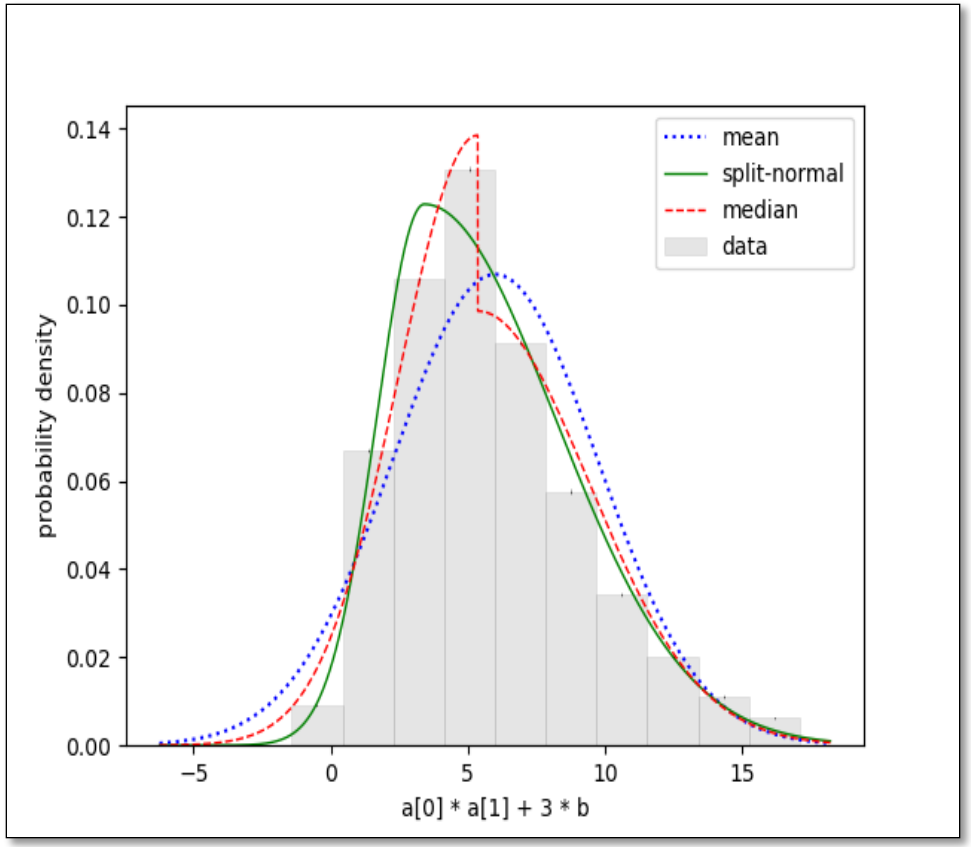
- Integration is a mathematical method for finding the area under a curve
- This is often equivalent to total accumulated change
- It is not always possible to find a solution analytically



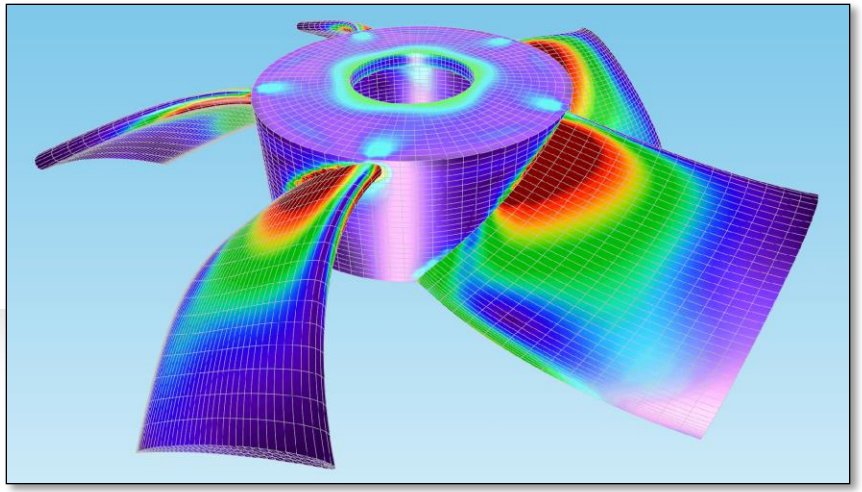
What is numerical integration

- Approximating a solution to an integral
- Many different techniques
- Monte Carlo picked due to ease of use and simplicity

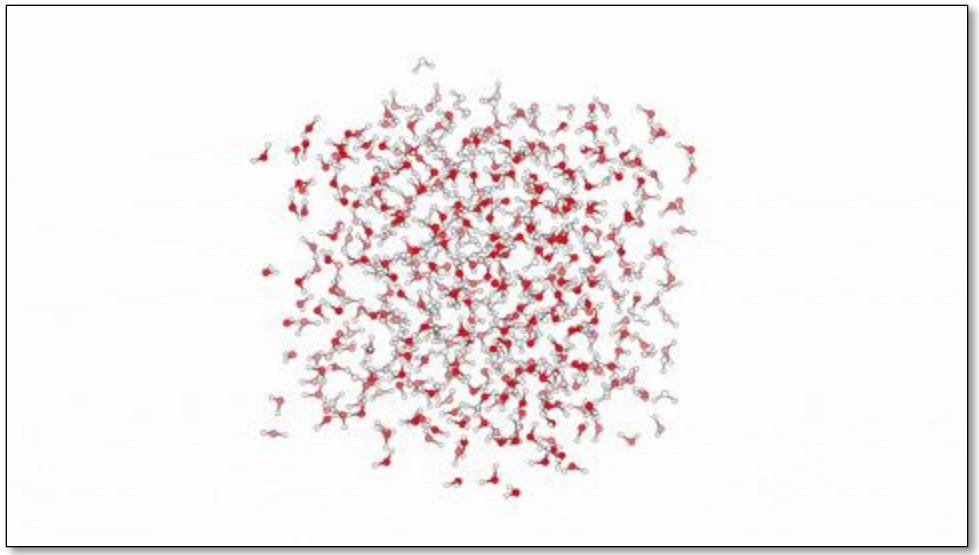




Applications

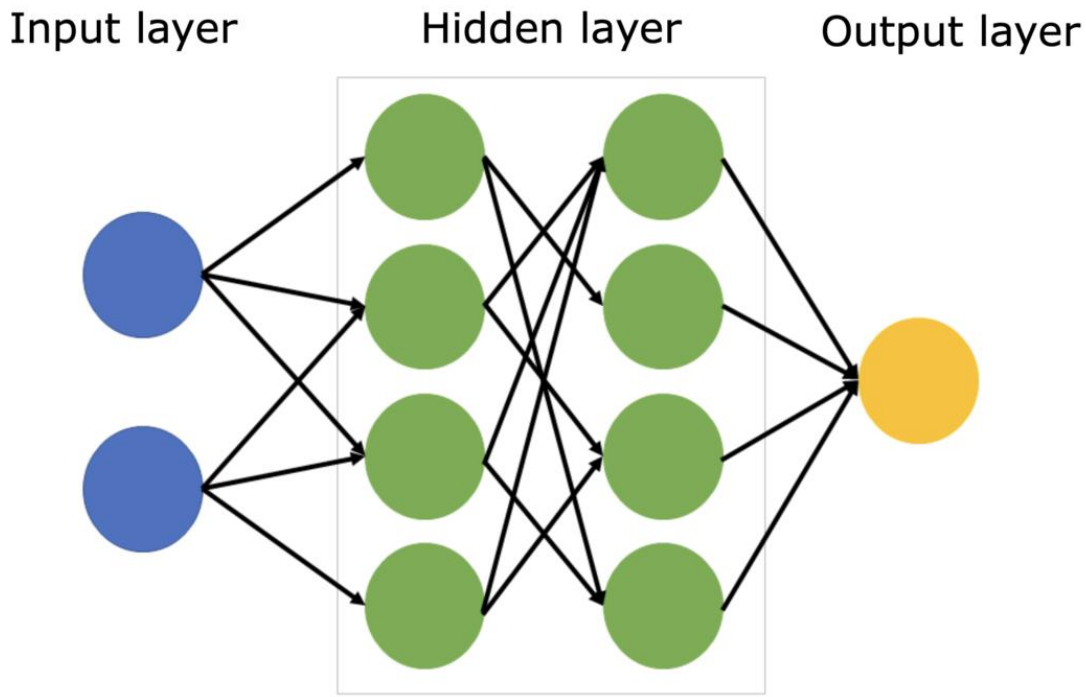


- Why is numerical integration important?
- Protein folding
- Rocketry
- Navier-Stokes equation
- Finite Element Analysis



What is a neural network

- Works like neurons in the brain
- Input data is processed by weights and biases
- Weights and biases adjusted using error to improve accuracy
- More and wider layers mean better performance but higher train time



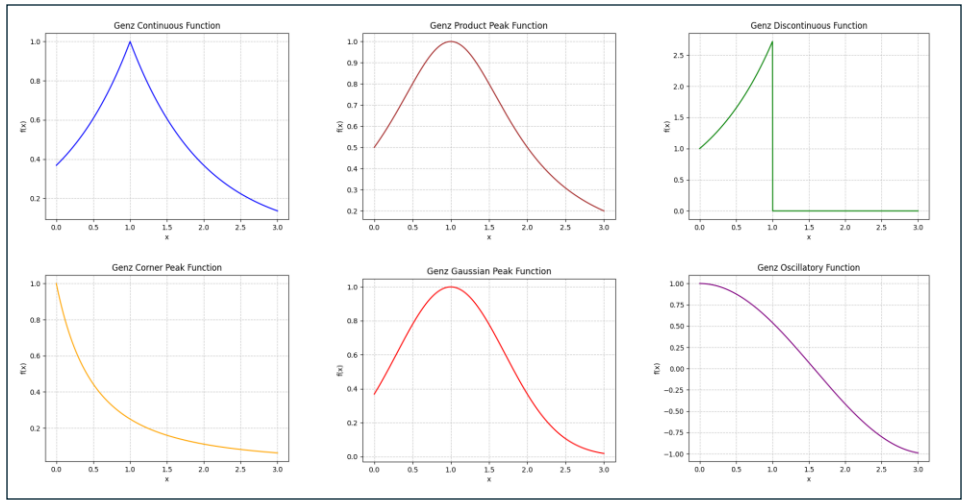
Artificial neural networks

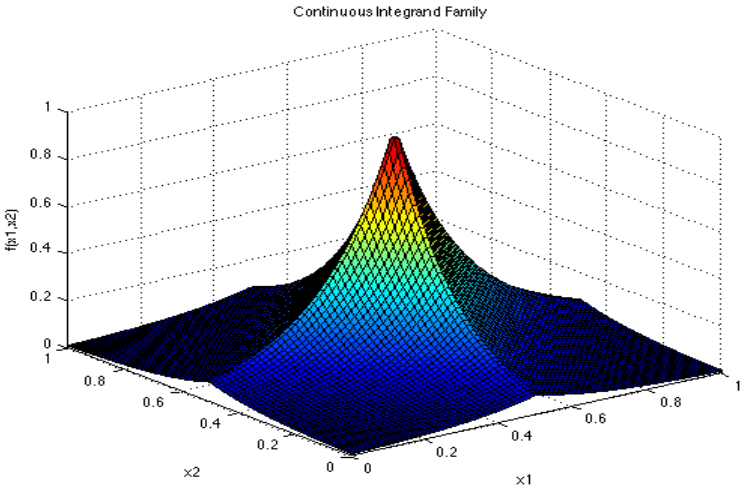
Testing if a neural network can integrate

- Neural network trained on an integral with an exact solution
- Network used to predict integral from 0 to 1
- Error was under 0.54 percent

Genz integral test functions

- Genz integral test functions are a set of six function families designed for testing numerical integration algorithms
- Extendable into d dimensions
- This project considered only one dimension





$$f(x) = \exp\left(-\sum_{i=1}^{d} a_i |x_I - u_i|\right)$$

for $d = 1$ $\quad f(x) = e^{-a_i|x_i - u_i|}$

```
def contin(x):
    return np.exp(-a * np.abs(x-u))
```

Designing a neural network

- Many design parameters have tradeoffs
- Others have no tradeoffs but still may affect accuracy
- 3 layer small network was chosen for speed and simplicity
- Hyperparameter tuning was not used

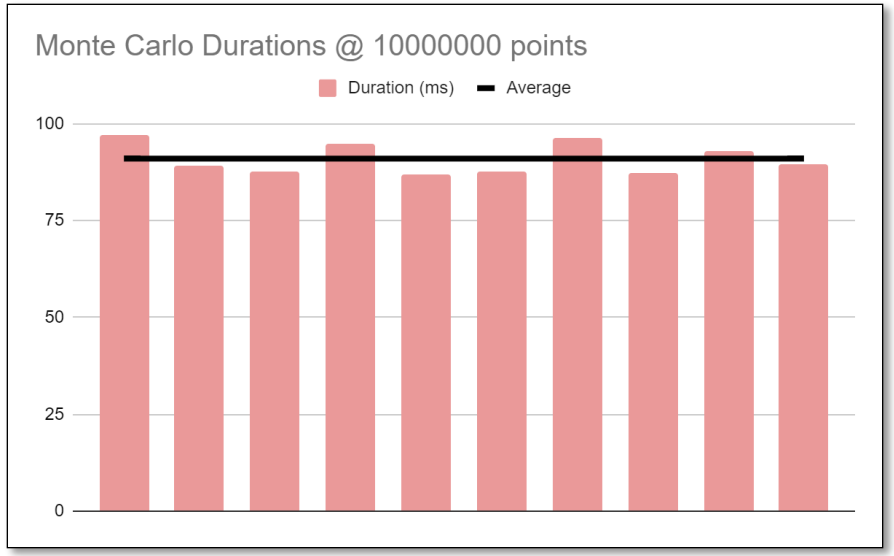
Comparison Setup

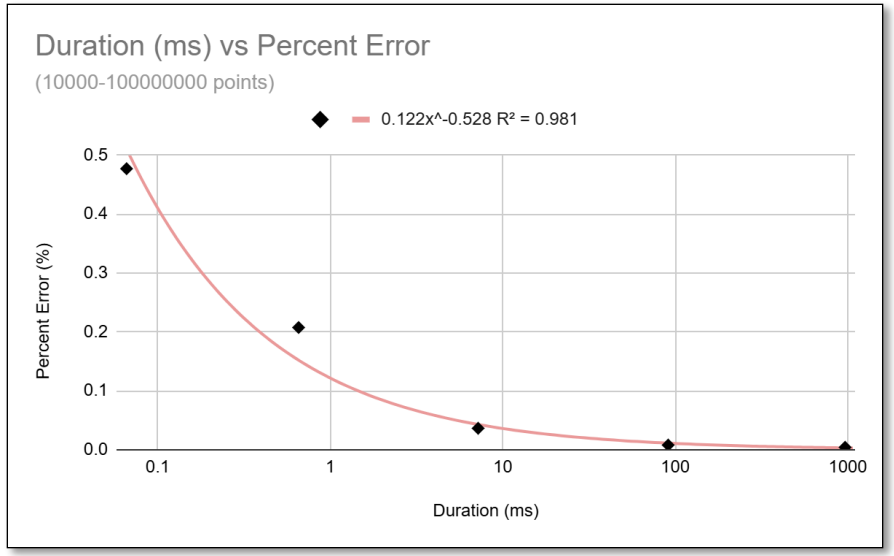
- Genz test functions and Monte Carlo integrator made in Python
- Keras3 used for neural network
- GPU and CPU neural networks tested
- Accuracy and time compared

Monte Carlo Method

- Estimates the value of an integral by averaging functions heights at random points and multiplying by the width of the domain
- Monte Carlo dependent on speed of evaluating the base function
- Effective and simple method

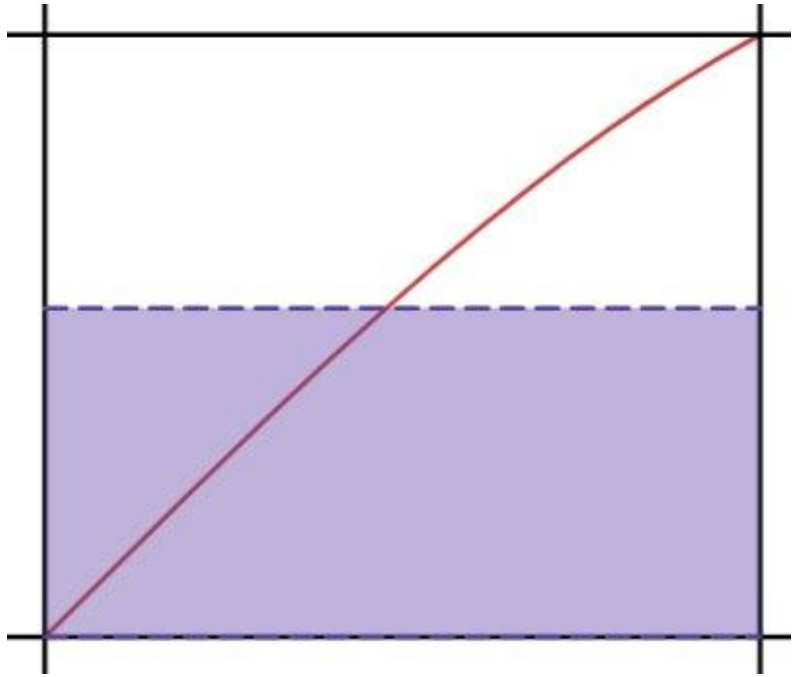
Continuous integrand family



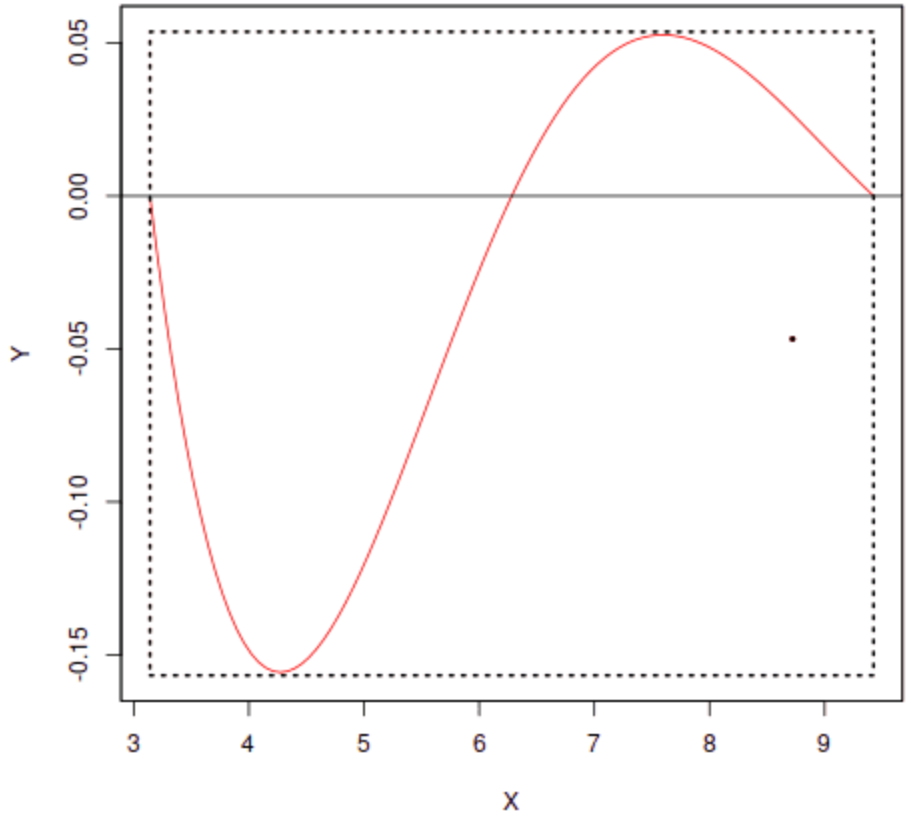


Monte Carlo Method

Mean-Value Monte Carlo Method

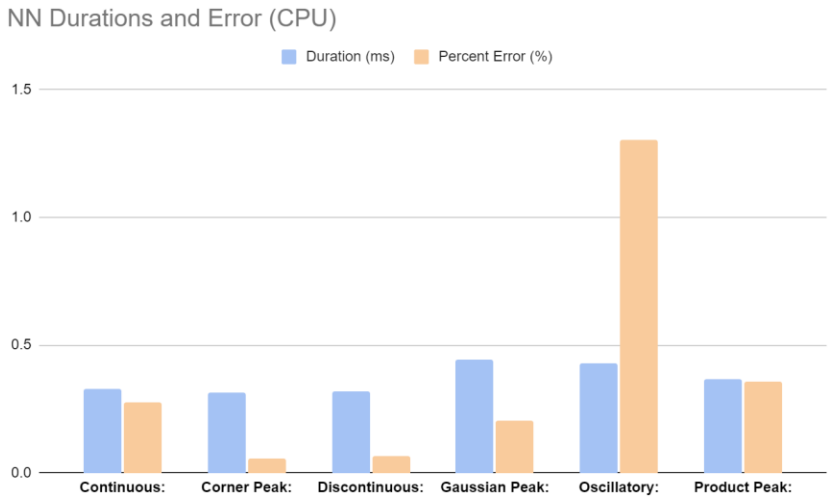


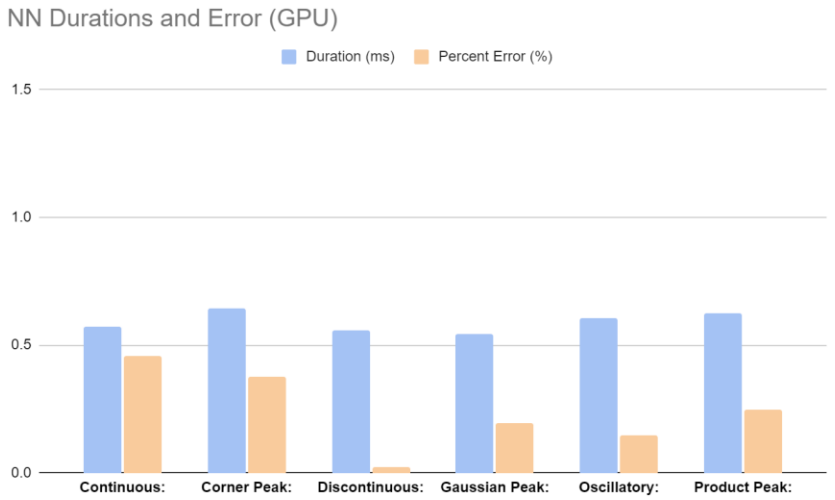
Hit or Miss Monte Carlo Visualized



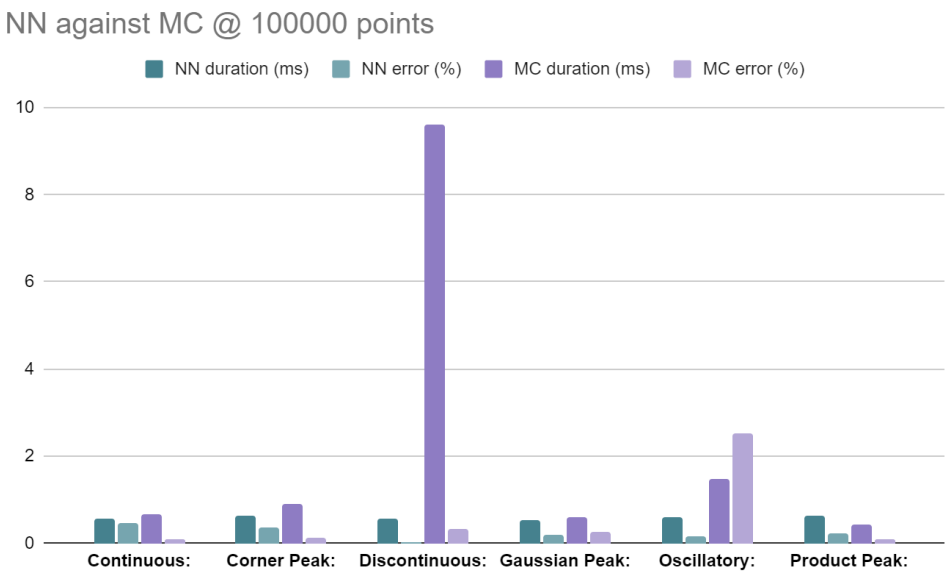
Neural Network

- Neural network speed independent of base function
- Oscillatory function less accurate with CPU (tested over multiple trials)





Comparison



- Neural network wins when function is slower to evaluate
- Monte Carlo is competitive for 1 dimension functions
- Network on average 35.5% faster and 121% more accurate on average excluding discontinuous
- For discontinuous network is 1623.5% faster and 1428.8% more accurate

Limitations

- Learning rate not adjusted
- No hyperparameter tuning
- Networks all use a fixed lower bound
- Only Monte Carlo tested
- Training cost and data-generation cost could be accounted for

Future work

- Use Raytune to find optimal hyperparameters
- Test additional computer configurations
- Compare vs VEGAS and Trapezoid rule
- Test multidimensional functions
- Test larger neural networks

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